Lokesh Muthuraj

Mechanical Engineering Student | CAD & Product Design | Automation & Emerging Technology

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PROFILE

Mechanical Engineering student at National Engineering College with hands-on experience in CAD, product design, thermal engineering, simulation, drones and automation. Skilled in SolidWorks, Fusion 360, AutoCAD, ANSYS and MATLAB, with project experience in autonomous disaster-response drones and finned double-pipe heat exchangers. Interested in product design, R&D, automation and innovation.

EDUCATION

B.E. Mechanical Engineering — **2024 – 2028**

National Engineering College, Tamil Nadu — CGPA: 7.84 (up to 4th semester)

TECHNICAL SKILLS

Mechanical	CAD · Product Design · Manufacturing · Heat Transfer · Thermal Engineering · Mechanical Design
Software	SolidWorks · Fusion 360 · AutoCAD · ANSYS · MATLAB
Programming	C · Python · HTML
Emerging Tech	Robotics · Drones · IoT · AI · Automation

PROJECTS

Disaster Response Drone for Remote Areas

Smart India Hackathon · SIH25047

- **Concept:** Autonomous heavy-lift drone for medical-supply and emergency-resource delivery in remote and inaccessible areas.
- **Systems:** Pixhawk 6X, NVIDIA Jetson, GPS, LiDAR, mmWave radar, thermal camera and ArduPilot; target payload 5+ kg.
- **Features:** Autonomous mission planning, obstacle detection/avoidance, thermal-based people detection, real-time monitoring and alerts.

Double Pipe Heat Exchanger with Fins (DPHE)

Simulation Using Modern Tools Project

- Designed and optimized a finned double-pipe heat exchanger for heat-transfer enhancement during milk pasteurization.
- Optimized fin height, thickness, spacing and number of fins using MATLAB; evaluated thermal performance using ANSYS simulation.

CERTIFICATIONS & WORKSHOPS

Certifications	CSWA – Certified SOLIDWORKS Associate (Dassault Systèmes) · Fusion for CAD · Manufacturing Documentation · QCAD Training · MATLAB Onramp · Machine Learning Onramp · Deep Learning Onramp
NPTEL	Principles of Metal Forming Technology (IIT Roorkee) · Characterization of Polymers, Elastomers and Composites (IIT Kharagpur)
Workshops	Drone Workshop (Mistral) · 3D Printing Technology (MIT Chennai) · Product Prototyping (PSG College of Technology) · Embedded System Architecture & ARM Processor (VIT Vellore)

HACKATHONS & TECHNICAL ACTIVITIES

Hackathons	Hack Odyssey 3.0 – National Level Hackathon · YHACK '26 – National Level 24-Hour Hackathon
Activities	Paper Presentation – Acumen'25 · Line Follower – Acumen'25 · Prompt Engineering – NeuraNexus 2K25 · Paper Presentation – Interofest 2K25

STRENGTHS & INTERESTS

Strengths	Problem Solving · Hands-on Prototyping · Team Collaboration · Quick Learner
Interests	CAD & Product Design · Manufacturing · Automation & Robotics · AI · Drones · Thermal Engineering